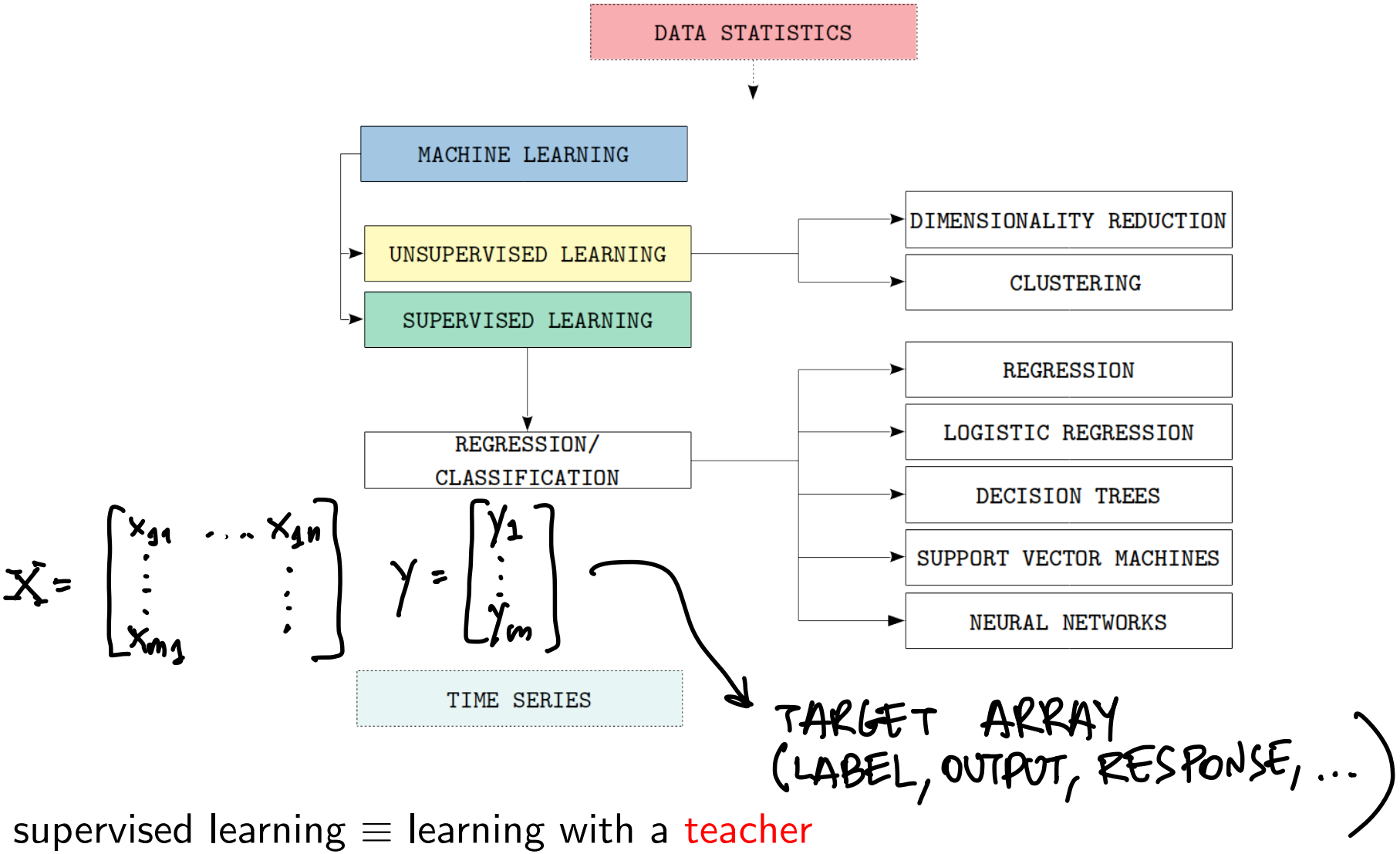
Overview



supervised learning $\equiv$ learning with a teacher

idea: observe many features-target pairs to reconstruct the function mapping features to target

supervised learning

- consider a process producing target y in correspondence to features $x \in \Re^n$ according to an unknown function

BUT POSTULATED

$$y = f(x) + \mathcal{E}$$

NOISE/ERROR

- supervised learning determines a model $m(\cdot)$ approximating $f(\cdot)$ on the basis of a set of known features-target pairs denoted as training set (TR)

INPUT - OUTPUT

$$TR = \{(x^i, y^i) : x^i \in \Re^n, y^i \in ?, i = 1, \ldots, m\}$$

HISTORICAL, AVAILABLE

- in the training phase the parameters of $m(\cdot)$ are tuned so that the model fits "as much as possible" TR

CARRIED OUT BY SOLVING AN OPTIMIZATION PROBLEM : A LOSS FUNCTION IS MINIMIZED

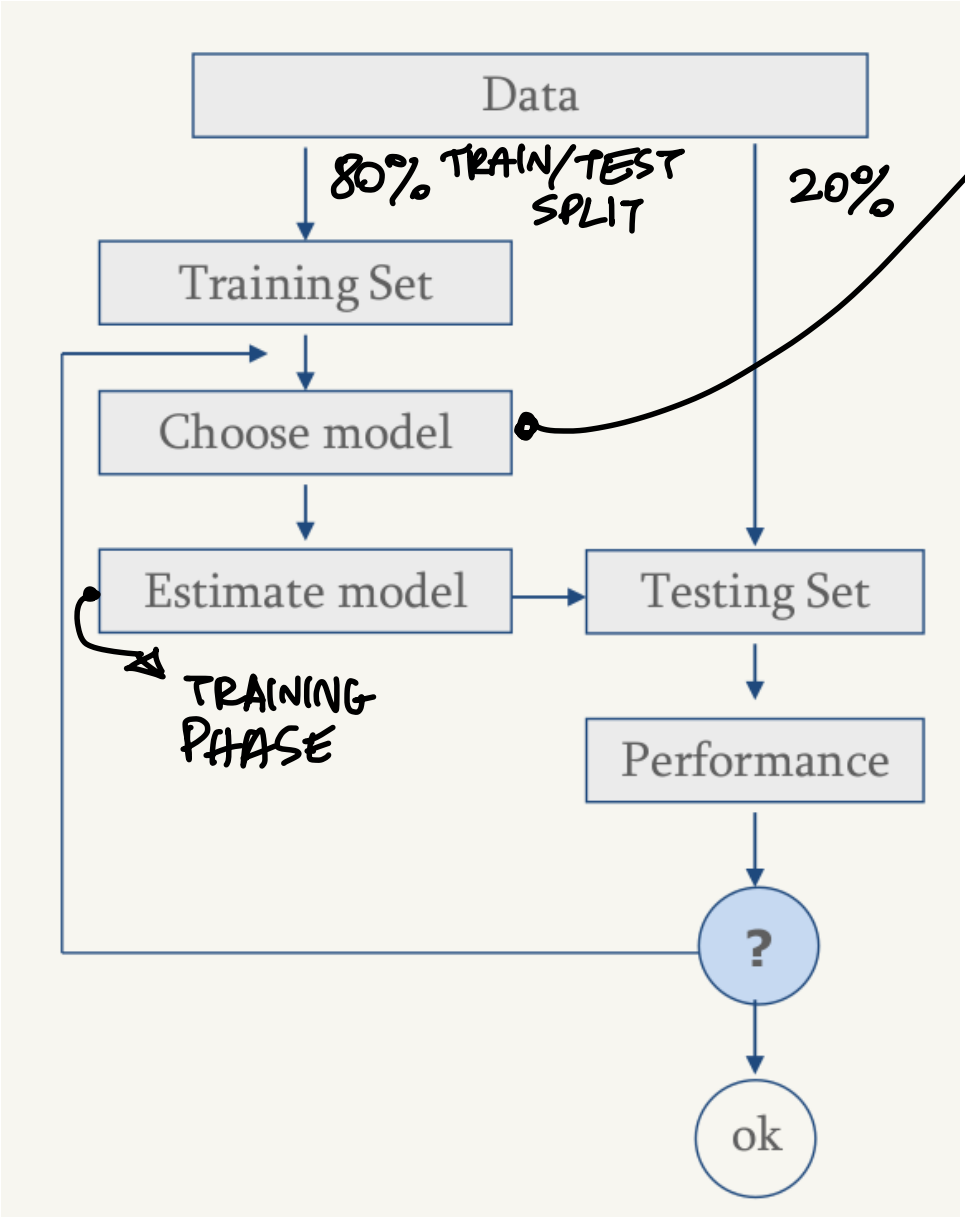
TRAINED

LOSS FUNCTION EXPRESSES THE DISCREPANCY BETWEEN OUTPUT OF $m(\cdot)$ AND y

- once trained the model is tested on a testing set (TS) of known features-target pairs not used in the training, and if "sufficiently good" it can be used to predict the target $\tilde{y}$ for unseen observations $\tilde{x}$ according to

$$\tilde{y} = m(\tilde{x})$$

supervised learning scheme



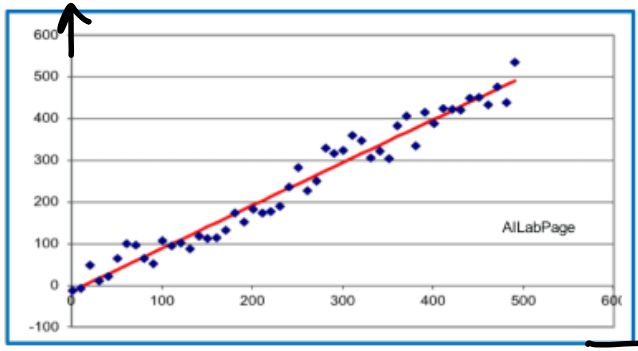
- generalization: capability of the trained model $m(\cdot)$ to correctly predict the target of unseen observations → NOT USED DURING THE TRAINING
- overfitting: when $m(\cdot)$ fits "too much" the training data and it is not able to make good predictions for unseen observations
  - GOOD PERFORMANCE ON TR
  - BAD PERFORMANCE ON TS

- the form of $f(\cdot)$ may be of any type
  - mathematical function
  - logic formula
  - the result of an algorithm
  - a black-box system
  - ...
- the form of $m(\cdot)$ may be of different types
  - mathematical function
  - a structured object (e.g., a tree, a neural network,...)
  - a set of rules
  - ...
- learning tasks categorized according to the domain of target y
  - y continuous $\Rightarrow$ regression
  - y discrete $\Rightarrow$ classification (binary or multi-class)

↳ CATEGORICAL

regression

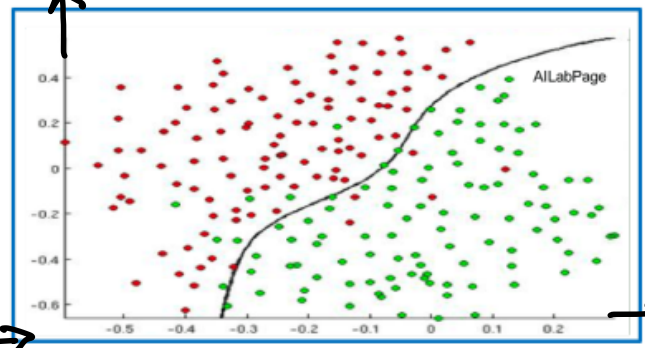
TARGET y



SINGLE FEATURE EXAMPLE

binary classification

FEAT. 2



2-FEATURES EXAMPLE

FEAT. 1

x_1 (FEAT. 1)

many issues in supervised learning...

- how to select the family of functions of model $m(\cdot)$
- how to manipulate data
- how to determine hyperparameters of $m(\cdot)$ (model parameters which cannot be tuned during the training phase and must be set a priori)
- which algorithm to adopt in the training phase
- how to split training and testing sets
- how to measure the performance of the trained model
- how to choose/control the model complexity (complex models fit training data well, simple models tend to better generalize)

... many alternative machine learning approaches

linear regression → REGRESSION TASK

linear regression fast and interpretable → M.L. MODELS USED BY HUMANS...

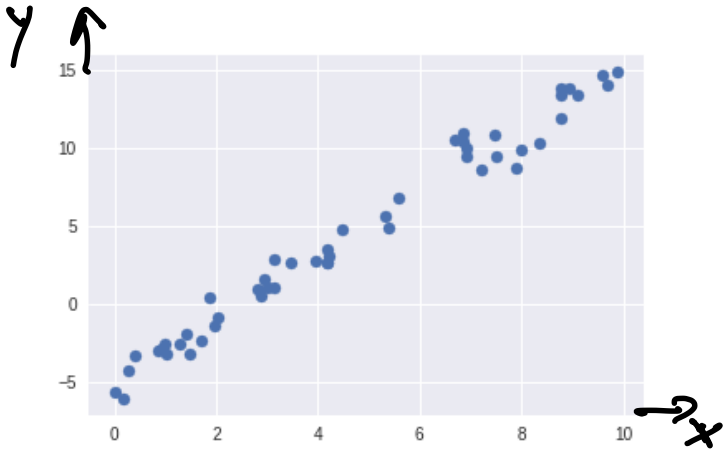
QUICKLY TRAINED
QUICK RESPONSE

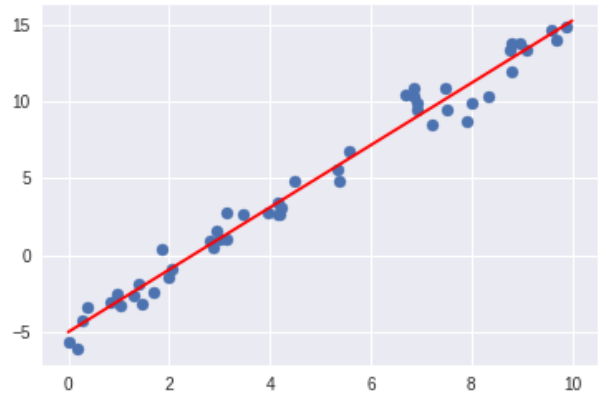
simple linear regression model (single feature)

$$y = \beta_0 + \beta_1 x \text{ (regression line)}$$

REGRESSION COEFFICIENTS

- $y \in \Re$ (regression task)
- β_0 intercept
- β_1 slope





training a simple linear regression consists in determinining $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the regression line fits ~~better~~ AS MUCH AS POSSIBLE the points IN TR

training simple linear regression

- given a TR of pairs (x^i, y^i) with $i = 1, \ldots, m$, find estimates $\hat{\beta}_0, \hat{\beta}_1$ such that

$$\hat{\beta}_0 + \hat{\beta}_1 x^i \approx y^i \quad \text{with } i = 1, \ldots, m$$

- given $\hat{\beta}_0, \hat{\beta}_1$, the prediction $\hat{y}^i$ for point x^i is

$$\hat{y}^i = \hat{\beta}_0 + \hat{\beta}_1 x^i$$

and the i-th error (residual) is computed as

$$e^i = |y^i - \hat{y}^i|$$

- $\hat{\beta}_0, \hat{\beta}_1$ are obtained by solving the following optimization problem where a loss function called residual sum of squares (RSS) is minimized

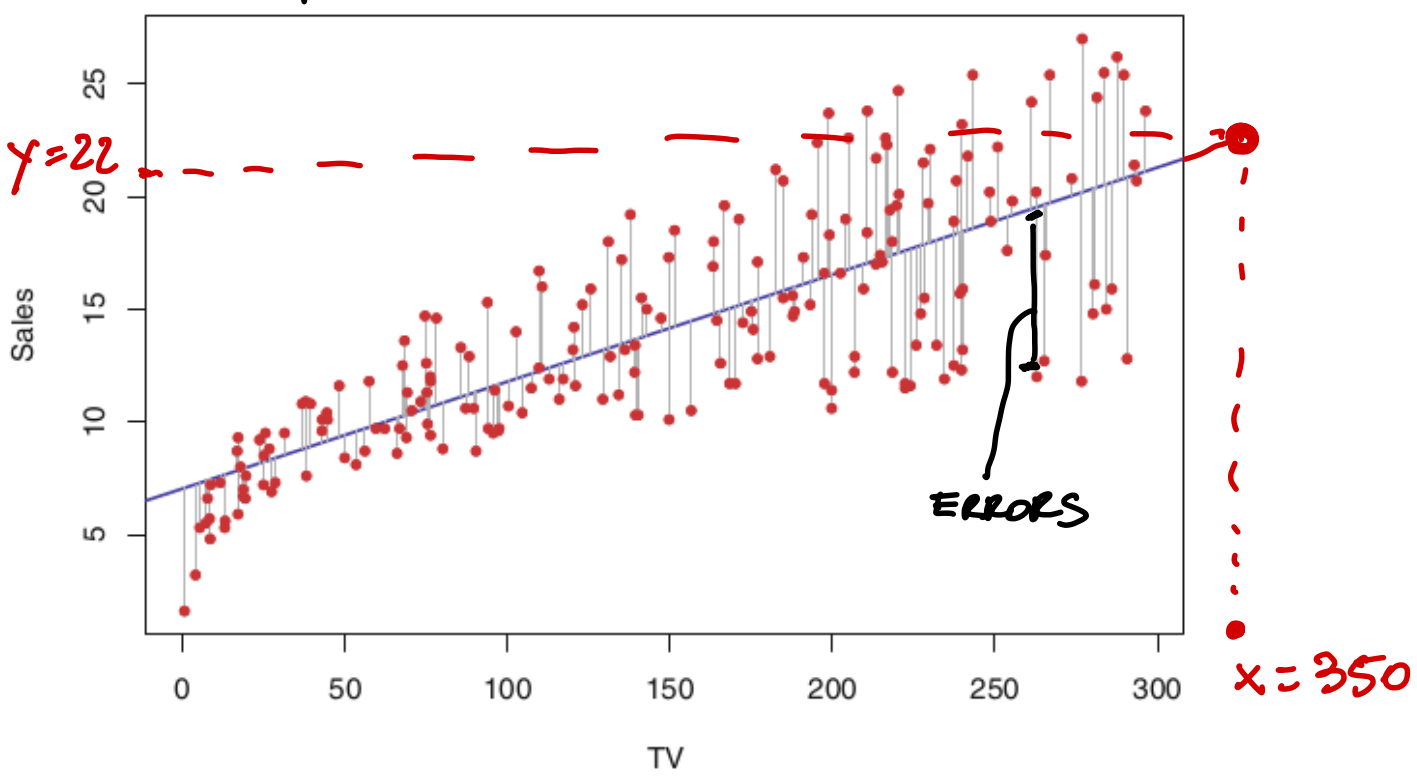
$$\min_{\beta_0, \beta_1 \in \Re} RSS = \sum_{i=1,m} (y^i - \beta_0 - \beta_1 x^i)^2 = \sum_{i=1,m} (y^i - \hat{y}^i)^2 = \sum_{i=1,m} (e^i)^2$$

- RSS is strictly convex $\Rightarrow$ the solution is unique:

$$\begin{cases} \hat{\beta}_1 = \dfrac{\sum_{i=1,m} (x^i - \bar{x})(y^i - \bar{y})}{\sum_{i=1,m} (x^i - \bar{x})} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

- prediction for an unseen $\tilde{x}$: $\tilde{y} = \hat{\beta}_0 + \hat{\beta}_1 \tilde{x}$

example: Sales of a product as a function of the TV advertisings over 200 different markets



$$\hat{\beta}_0 = 7.03, \hat{\beta}_1 = 0.0475 \Rightarrow \texttt{Sales} = 7.03 + 0.0475 \ \texttt{TV}$$

by spending 1000 more euro in TV advertisings we sell 47.5 more units of product!

multivariate linear regression: predictors (features) are n

MULTIPLE SLOPE COEFFICIENTS

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

- given $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_n$, the prediction $\hat{y}^i$ for point $x^i \in \Re^n$ is

$$\hat{y}^i = \hat{\beta}_0 + \hat{\beta}_1 x_1^i + \hat{\beta}_2 x_2^i + \cdots + \hat{\beta}_n x_n^i$$

- $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_n$ are obtained by solving

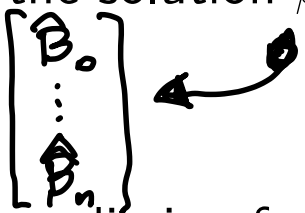
MODEL OUTPUT

$$\min_{\beta_0,\beta_1,\beta_2,\ldots,\beta_n} RSS = \sum_{i=1,m} (y^i - \beta_0 - \beta_1 x_1^i - \beta_2 x_2^i - \cdots - \beta_n x_n^i)^2 =$$

REAL OUTPUT

$$= \sum_{i=1,m} (y^i - \hat{y}^i)^2 = \sum_{i=1,m} (e^i)^2$$

- the solution $\hat{\beta} \in \Re^{n+1}$ can be computed as



$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$X^{m \times (n+1)}$ = FEATURE MATRIX

$y^{m \times 1}$ = TARGET ARRAY

- prediction for an unseen $\tilde{x}$: $\tilde{y} = \hat{\beta}_0 + \hat{\beta}_1 \tilde{x}_1 + \hat{\beta}_2 \tilde{x}_2 + \cdots + \hat{\beta}_n \tilde{x}_n$

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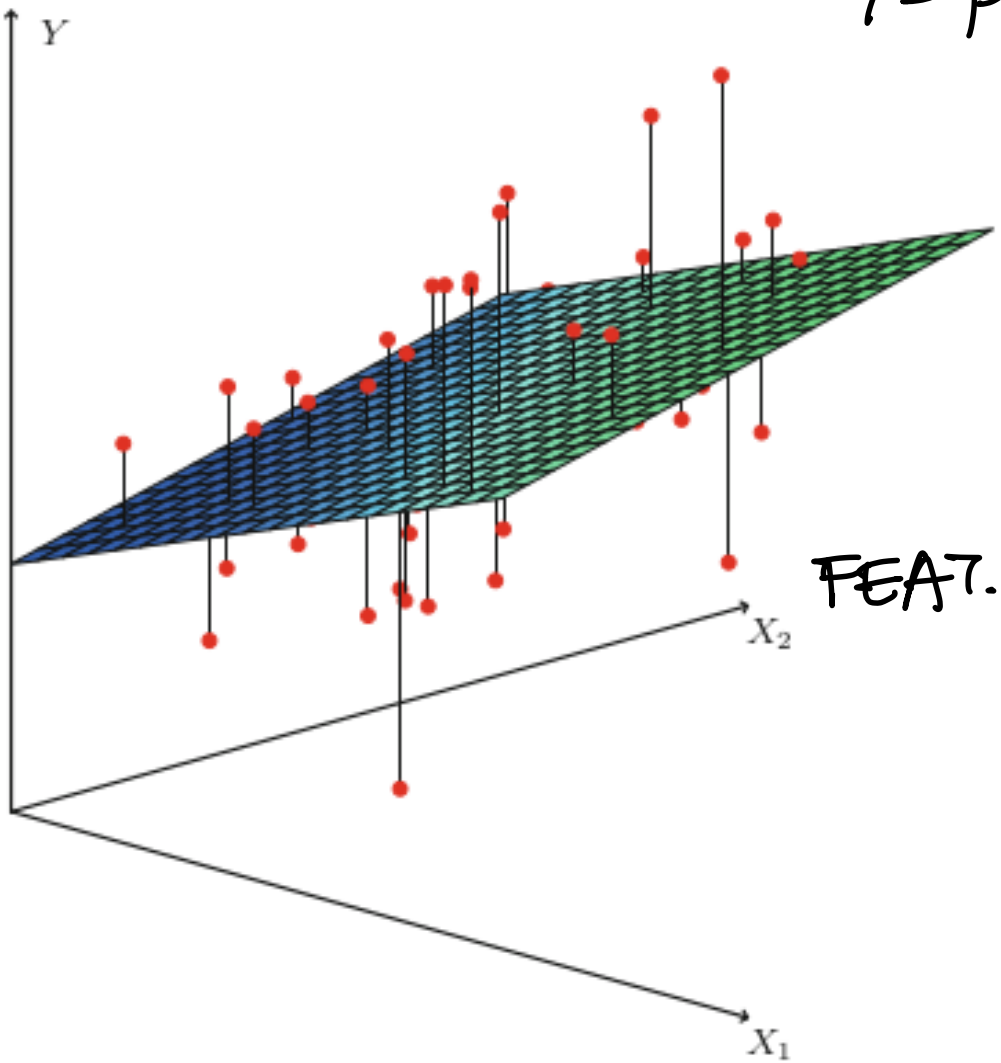
a 2-dimensional linear regression

$n = 2$

$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

$\Downarrow$ TRAINING

$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$



FEAT. 2

FEAT. 1

how to measure the regression accuracy (on the training or testing set)?

- residual sum of squares (RSS)

$$RSS = \sum_{i=1,m} (y^i - \hat{y}^i)^2$$

- IT DEPENDS ON m
- IT DEPENDS ON THE Y SCALE

MODEL OUTPUT

REAL OUTPUT

THE SCALE OF RSS IS NOT THE SCALE OF Y

- mean square error (MSE)

IT DOES NOT DEPEND ON m

$$MSE = \frac{1}{m} \sum_{i=1,m} (y^i - \hat{y}^i)^2$$

(root mean square error ($RMSE$) $= \sqrt{MSE}$)

THE SAME SCALE OF Y

- coefficient of determination (R^2)

total sum of squares $TSS = \sum_{i=1,m} (y^i - \bar{y})^2$ $\quad \bar{y} = \frac{1}{m} \sum_{i=1,m} y^i$

INTRINSIC VARIABILITY OF THE DATASET

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{TSS - RSS}{TSS}$$

$R^2 \in [0,1]$

ERROR OF THE MODEL

VARIABILITY THAT MODEL CANNOT EXPRESS